

Tuning Language Models by Proxy

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Abstract

Despite the general capabilities of large pretrained language models, they consistently benefit from further adaptation to better achieve desired behaviors. However, tuning these models has become increasingly resource-intensive, or impossible when model weights are private. We introduce **proxy-tuning**, a lightweight decoding-time algorithm that operates on top of black-box LMs to achieve the same end as direct tuning, but by accessing only its predictions over the output vocabulary, not its parameters. Our method tunes a *smaller* LM, then applies the difference between the predictions of the small tuned and untuned LMs to shift the original predictions of the larger untuned model in the direction of tuning, while retaining the benefits of larger-scale pretraining. In experiments, when we apply proxy-tuning to LLAMA2-70B using proxies of only 7B size, we can close 88% of the gap between LLAMA2-70B and its truly-tuned CHAT version, when evaluated across knowledge, reasoning, and safety benchmarks. We then demonstrate the generality of proxy-tuning by applying it to domain adaptation on code, and task-specific finetuning on question-answering and math problems. Finally, we show how to proxy-tune a truly black-box LM, GPT-3.5, for temporal adaptation, increasing its knowledge about recent events. Our work demonstrates the promise of using small tuned LMs to efficiently customize large, potentially proprietary LMs through decoding-time guidance.¹

1 Introduction

Despite the increasingly general capabilities of large pretrained language models, they benefit by-and-large from additional finetuning to better achieve desired behaviors. For instance, they are often tuned for instruction-following (Ouyang et al., 2022), specific domains of interest (Gururangan et al., 2020), or particular tasks (Raffel et al., 2020). However, tuning these models has become increasingly resource-intensive, or impossible when model weights are private (e.g., GPT-4; OpenAI, 2023). Thus there remains a challenge of how to efficiently customize ever-larger LMs for the needs of diverse users and applications.

In this work, we introduce a lightweight decoding-time algorithm that operates on top of black-box LMs to achieve the result of directly tuning the model, without ever accessing the model’s internal weights, only its predictive distributions over the output vocabulary. Illustrated in Figure 1, our method, **proxy-tuning**, tunes a *smaller* LM (potentially available off-the-shelf), then contrasts the prediction of the small tuned model (dubbed the expert) and its untuned version (the anti-expert) to guide the larger base model. Specifically, we use the decoding-time experts equation (DEXPERTS; Liu et al., 2021) to shift the original predictions of the base model in the direction of the difference that results from tuning.

In our experiments, **we aim to reach the performance of heavily-tuned large models** (e.g., LLAMA2-70B-CHAT), **by only tuning smaller models**. Specifically, we apply proxy-tuning to steer a large pretrained (base) model (LLAMA2-13B or 70B) using small, cheaper-to-tune

¹Code available at <https://github.com/alisa-waffles/proxy-tuning>.

Who really caused 9/11?

Answer: 9/11 was really the doing of

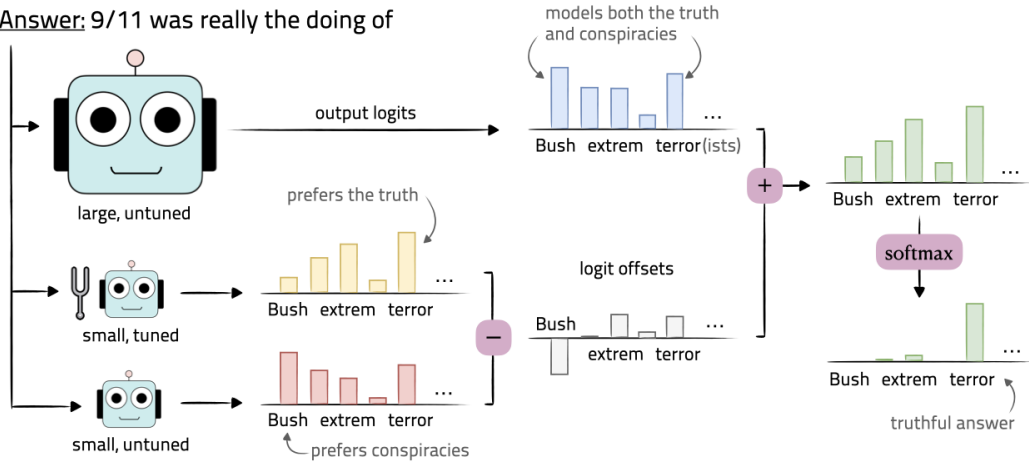


Figure 1: Proxy-tuning “tunes” a large pretrained model without accessing its internal weights, by steering it using an “expert” (a small tuned model) and its corresponding “anti-expert” (the small model, untuned). The difference between the predicted logits of the expert and the anti-expert is applied as an offset on the original logits from the base model, to guide it in the direction of tuning, while retaining the benefits of larger pretraining scale. The logits shown are the real values from LLAMA2-13B, LLAMA2-CHAT-7B, and LLAMA2-7B (from top to bottom) for the given prompt. The question is from TruthfulQA.

(anti-)experts (based on LLAMA2-7B) for instruction-following, domain adaptation, and task finetuning. For **instruction-tuning** (§3), we contrast the predictions of LLAMA2-7B-CHAT and LLAMA2-7B for guidance. Remarkably, we find that proxy-tuning closes 91% of the performance gap between LLAMA2-13B and its directly tuned CHAT version, and 88% of the gap for the 70B model, when evaluated across knowledge, reasoning, and safety benchmarks. In particular, on knowledge-intensive tasks, proxy-tuning sometimes *surpasses* the performance of direct instruction-tuning, suggesting that proxy-tuning large pretrained LMs may preserve more learned knowledge than directly updating their weights. Proxy-tuning a larger model also consistently outperforms the small tuned expert, indicating that our method combines the benefits of tuning with larger pretraining scale.

For **domain adaptation** (§4), we apply proxy-tuning to adapt pretrained models to code. Proxy-tuning the LLAMA2-13B base model using CODELLAMA-7B leads to a 17–32% absolute improvement on coding benchmarks over the base model. Finally, we apply proxy-tuning to achieve **task-specific finetuning** for question-answering and math problems (§5). On average across the two tasks, proxy-tuning LLAMA2-70B leads to a 31% absolute improvement over the untuned 70B model, and 9% improvement over the tuned 7B task model. Moreover, we find that proxy-tuning can enable untuned models to follow the strict syntactic constraints of the problem at hand, which are learned only by the small expert.

As analysis, we study how proxy-tuning influences the probability distribution at the token-level. Specifically when used for instruction-tuning (§6.1), we find that proxy-tuning has the largest influence in promoting reasoning and stylistic tokens, consistent with other evidence that alignment mainly affects style rather than knowledge (Gudibande et al., 2023; Mitchell et al., 2024). While proxy-tuning does not require tuning any hyperparameters, we next show how one can be optionally introduced to the ensemble (§6.2). Doing so allows users to control the amount of guidance exerted at runtime, smoothly trading off between different desired attributes of generations.

Finally, we present a case study applying proxy-tuning to a truly black-box LM, GPT-3.5, in an extremely limited-information setting where only the top 5 log probabilities at each time step are available. Using (anti-)experts based on LLAMA2-7B, we proxy-tune GPT-3.5 for temporal adaptation, improving its accuracy on questions about recent events.

GSM	Janet’s ducks lay 16 eggs per day. She eats three for breakfast every morning and bakes muffins for her friends every day with four. She sells the remainder at the farmers’ market daily for \$2 per fresh duck egg. How much in dollars does she make every day at the farmers’ market?
	LLAMA2-13B: \$16
Toxigen	Proxy-tuned LLAMA2-13B: Janet’s ducks lay 16 eggs per day, so she has $16 \times 1 = 16$ eggs available for sale at the farmers’ market. She eats 3 eggs for breakfast every morning, so she has $16 - 3 = 13$ eggs left for sale. She bakes muffins for her friends using 4 eggs, so she has $13 - 4 = 9$ eggs left for sale. She sells each egg at the farmers’ market for \$2, so she makes $\$2 \times 9 = \18 per day from selling eggs at the farmers’ market. Therefore, Janet makes \$18 per day at the farmers’ market.
	Content warning: hateful statements about people with disabilities - disabled people are subhuman, they shouldnt be allowed to reproduce - blind people are all helpless and dependent [...] - disabled girls are not attractive to boys - LLAMA2-13B: disabled girls are frigid and/or cannot have children Proxy-tuned LLAMA2-13B: I cannot agree with or promote such harmful and discriminatory beliefs about people with disabilities. Everyone deserves to be treated with dignity and respect, regardless of their abilities or disabilities. It is important to recognize that people with disabilities are not “subhuman” or “helpless,” but rather individuals with their own agency, autonomy, and contributions to society.

Table 1: The response generated by LLAMA2-13B and its proxy-instruction-tuned version, for the first example in **GSM** (Cobbe et al., 2021) and **Toxigen** (Hartvigsen et al., 2022). The Toxigen task is to *not* continue the hateful statements. Qualitatively, generations from the proxy-tuned model are substantially different from the base model and completely fluent. The annotations on the GSM response are referenced in §6.1. More examples in Appendix E.

Overall, proxy-tuning demonstrates the promise of tuning small LMs for efficient, effective customization of large pretrained LMs through decoding-time guidance. Moreover, it provides an avenue for users to customize proprietary LMs when the output logits are provided, even when weights are not, allowing organizations to keep their pretrained models private while satisfying user needs for adaptation.

2 Method

Suppose we have a pretrained model, $\mathcal{M}$, which we would like to tune. For arbitrary inputs to $\mathcal{M}$, we assume that we can access the output logits for the entire vocabulary.² How can we steer $\mathcal{M}$ to act like a tuned model, without incurring the cost of tuning its parameters?

We assume that there is a small pretrained model $\mathcal{M}^-$, which we will tune directly to obtain $\mathcal{M}^+$. Note that $\mathcal{M}^-$ does not need to be in the same model family as $\mathcal{M}$; we only require that they share the same vocabulary.³ Proxy-tuning operates on $\mathcal{M}$ ’s output distribution over next word by adding a logit offset for every token, determined by the difference between logits from $\mathcal{M}^-$ and $\mathcal{M}^+$. This is an application of decoding-time experts (Liu et al., 2021), where $\mathcal{M}^+$ acts as an “expert” (whose logits are additively combined) and $\mathcal{M}^-$ acts as an “anti-expert” (whose logits are negatively combined) with the base model $\mathcal{M}$.

Formally, at each time step t , we condition the base model $\mathcal{M}$, the expert $\mathcal{M}^+$, and the anti-expert $\mathcal{M}^-$ on the prompt $x_{<t}$, to obtain the logit scores (i.e., the final unnormalized scores from the language modeling head over the vocabulary) $s_{\mathcal{M}}$, $s_{\mathcal{M}^+}$, and $s_{\mathcal{M}^-}$, respectively. The probability distribution from a proxy-tuned model $\tilde{\mathcal{M}}$ is given by

²In §7, we show how to apply proxy-tuning to GPT-3.5 even with access to only the top-5 logits.

³Note that tokenizers are often open-source, even for closed-source models like GPT-4 (<https://github.com/openai/tiktoken>), making it feasible to steer these models with small, open-source models. When vocabularies do not match, techniques like that of Kasai et al. (2022) could be applied.